



PREMIUM-4 STAIRCASE TOWER

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Painted Finish
Anti-Skid Dimple Surface**Prime Steeltech (I) Pvt. Ltd.**
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IS 1161:2014

IS 3696 (Pt.1):1987

IS 4014 (Pt.1&2):1967

BS EN 12811-1

BS 1139-3

OSHA 29 CFR 1926.451

ISO 9001:2015

The **PREMIUM Cuplock Staircase Tower** is Prime Steeltech's standard 4-leg staircase access system, engineered for reliable, tool-free scaffolding access on construction and industrial sites. It features an **Anti-Skid Dimple** surface on all step arms and landing platforms, giving secure grip in wet and dusty site conditions while carrying heavy foot traffic and material loads.

Full-height guard rails at 940 mm and mid-rails at 470 mm run along both stair sides, 150 mm toe boards protect all three open landing edges, and adjustable base jacks with 450 mm of travel handle uneven ground. Anchor fastening provision on every tie bar secures the tower to the RCC structure at every level.

The 4-leg PREMIUM configuration spans a 2,500 × 2,000 mm grid and supports stair flights up to 60 m in platform height. Compatible with all major scaffolding systems and compliant with IS, BS EN and OSHA standards, the PREMIUM is built for high-rise, industrial and infrastructure projects.

Designed for compatibility with Ringlock and Cuplock scaffolding systems

GRID SIZE**2,500 × 2,000 mm****SYSTEM****Cuplock / Ringlock****STEEL GRADE****M.S./H.R. IS 2062****FINISH****Red-Oxide Primer + Blue Enamel****STAIR SURFACE****Anti-Skid Dimple**

Overall Tower Geometry

GRID SIZE (L × W)

2,500 × 2,000 mm

PLATFORM HEIGHT

Up to 60 m

TOWER OVERALL HEIGHT

Platform height + 1,500 mm (top rail)

CONFIGURATION

PREMIUM-4, 4-leg cuplock

Stair Geometry

Parameter	Specification
Stair Width (clear)	800 mm (min. 600 mm per EN 12811-1 Cl. 5.3; min. 457 mm per OSHA 29 CFR 1926.452(v))
Stair Type	Step arm with Anti-Skid Dimple
Stair Material	G.P. / M.S. sheet with Anti-Skid Dimple
Side Railing Height	940 mm
Guard Rail (Mid Rail)	470 mm
Railing — Both Sides	Yes
Step Rise	230 mm (Class-B per EN 12811-1; BS 1139-3)
Tread	300 mm (Class-B per EN 12811-1; BS 1139-3)
Staircase Angle	49° (EN 12811-1: 30–55°; OSHA 1926.451: 40–60°; BS 1139-3)
Step Arm UDL (Design Load)	450 kg/m² (≥ 1.0 kN/m² per EN 12811-1 Cl. 6.2.4; IS 3696-1)

Landing Platform — Quick Reference

LANDING WIDTH 2,000 mm	LANDING DEPTH 600 mm	TOE GUARDS 3 sides · 150 mm
LANDING FIXING Hook-on, secure	PLATFORM UDL 450 kg/m²	SURFACE Anti-Skid Dimple

Structural Framing

Component	Specification
Cuplock Vertical (Standard)	48.3 OD × M.S. IS 1161 Grade YST210
Top Cup	Drop-forged steel
Bottom Cup	Pressed metal
Cup Spacing	500 mm centers
Cuplock Ledger	48.3 OD M.S. pipe, forged ledger blades each end
Spigot Pin	38 mm OD pipe + 2 nuts & bolts, 250 mm long (Mandatory Above 30 m Height)
Bracing Pipe	48.3 OD M.S. pipe, all 4 sides
Couplers & Clamps	Heavy-duty forged; IS 2750 / EN-74 compliant

Base & Anchorage

Component	Specification
Adjustable Base Jack	Spindle 38 mm OD · total length 600 mm · adjustable 450 mm · base plate 150 × 150 × 6 mm M.S.
Tie Bar	48 mm OD M.S. pipe + welded holding plate 150 x 150 x 6 mm · anchor fastening provision · secures tower to RCC / walls

All dimensions in mm unless noted. Weights subject to ±3.0% tolerance. Platform height configurable up to 60 m; structural design required for heights exceeding 10 m.

Applicable Standards & Design Codes

Standard	Scope / Relevance
IS 1161:2014	Steel tubes for structural purposes — OD, wall thickness & mechanical properties (YST210/240/310) for all M.S. tubes.
IS 2750:1964	Steel scaffolding — fittings. Drop-forged and pressed couplers and clamps.
IS 3696 (Pt.1):1987	Safety code for scaffolds & ladders — stairway design, railings, toe-boards, platform protection.
IS 4014 (Pt.1):1967	Steel tubular scaffolding — materials. Component dimensions, grades, workmanship.
IS 4014 (Pt.2):1967	Steel tubular scaffolding — safety regs. Stairway access >4 m; handrails both sides; tie-bar spacing; bracing.
IS 2062	M.S. / H.R. sheet standard
BS EN 12811-1:2003	Temporary works equipment — scaffolds. Load class, stair width ≥ 600 mm, angle 30–55°.
BS 1139-3	Metal scaffolding — stairway towers. Stair geometry, Class-B rise/tread, tower configuration.
OSHA 29 CFR 1926.451	Scaffolds — construction industry (USA). Stair width min. 457 mm, angle 40–60°, fall protection > 3.05 m.
ISO 9001:2015	Quality management systems — Prime Steeltech QMS certification.

Key Features & Applications

Cuplock System

Tool-free assembly — cup locks up to 4 ledgers per node simultaneously.

Full Fall Protection

Guard rail @ 940 mm & mid-rail @ 470 mm; both sides of stair throughout tower height.

Height — Up to 60 m

Platform configurable up to 60 m; tie bars mandatory per IS 4014-2.

Platform UDL 450 kg/m²

Design load 4.41 kN/m²; exceeds EN 12811-1 Cl. 6.2.4 minimum (1.0 kN/m²).

Anti-Skid Surface

M.S./H.R. dimple-pressed treads and landings; secure grip in wet and dusty conditions.

Toe Guards — 3 Sides

150 mm toe boards on all open landing sides; prevents falling objects.

Tie Bar to RCC/Walls

48 mm OD tie bar with anchor fastening (150 x 150 x 6 mm); anchors tower to structure.

Design & Certification

Structural drawings, load calculations, and third-party certification included.

Typical Applications

- High-rise residential & commercial construction
- Infrastructure: bridges, flyovers, metro viaducts
- Industrial plants: refineries, petrochemical, power, cement
- Nuclear energy & defense sector structures
- Oil & gas platforms, jetty structures
- Demolition & decommissioning works
- Façade installation, cladding & glazing works
- Maintenance & renovation of existing structures
- Shipyards & dockyard construction
- Film production & event scaffolding (elevated platforms)
- Temporary public access: grandstands, events
- Aircraft maintenance hangar scaffolding

SAFETY NOTICE — Staircase towers must be designed, erected, and inspected by competent persons only, in accordance with IS 3696 Part 1, IS 4014 Parts 1 & 2, and the . Guard rails at 1,100 mm and mid-rails at 700 mm must be in place, with toe boards fitted, before personnel are permitted access at height.

Notes & Disclaimers

1. Staircase towers must be designed, erected, and inspected by competent persons in accordance with IS 3696 Part 1 and the IS 4014 Parts 1 & 2.
2. Tie bars must be fixed to the RCC or structural frame / walls at intervals per project-specific structural design. Maximum tie spacing: 4 m vertical, 6 m horizontal (IS 4014 Part 2 Cl. 12). Ties are mandatory once height exceeds 3× the least base dimension.
3. Adjustable base jack spindle: 38 mm OD pipe; total length 600 mm; usable / adjustable length 450 mm. Must not be extended beyond the usable length; use sole boards on soft or unpaved ground (IS 4014 Part 1).
4. Anti-skid dimple treads must be inspected periodically; worn or damaged treads must be replaced immediately.
5. Platform UDL of 450 kg/m² (4.41 kN/m²) is the design value. Site loadings must be confirmed by a structural engineer; do not exceed this value in service.
6. This datasheet is for information and quotation reference only. Structural calculations and certified drawings are prepared separately and are project-specific. Contact Prime Steeltech for project-specific design.

Contact & Further Information

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